

Izuma Device Management

Chip-to-Cloud IoT at Scale

Building custom IoT device management infrastructure takes months to years and diverts engineering from product differentiation. Izuma Device Management compresses that timeline from years to a day — a proven platform with over a decade of evolution (originally Arm Pelion / mbed Cloud) that handles secure connectivity, firmware updates, and fleet operations from ultra-constrained MCUs to full edge servers. **SOC 2 certified.**

Core Capabilities

Secure Device Connectivity

LwM2M/CoAP with DTLS. X.509 mTLS. Hardware root of trust (TPM 2.0, TrustZone, PARSEC). Protocol translation for BLE, Zigbee, Z-Wave, BACnet, Modbus. Outbound HTTPS only.

Firmware Updates (FOTA)

One-shot, continuous, and mesh campaigns. Delta updates. Encrypted and signed images. Resume after disconnect. Scales from MCU binaries to multi-GB images.

Fleet Management

Device Directory scales from dozens to millions. Filter, group, suspend, bulk-operate. First-to-claim enrollment. Custom attributes.

Edge Computing & Monitoring

True Kubernetes (not a fork). Offline container operation. SD-WAN with firewall traversal. Device Sentry threshold-based incidents. Remote log collection and terminal.

Device Spectrum

Device Class	Stack
Constrained MCUs	Izuma Connect (Lite)
Mainstream MCUs / MPUs	Izuma Connect
Rich nodes (Linux)	Izuma Connect
Gateways	Izuma Edge
Edge servers (k8s)	Izuma Edge

Deployment Options

Model	Infrastructure
Fully Managed	SaaS on AWS (NoOps, 24/7)
Any Cloud	AWS, Azure, GKE + Izuma support
Bare Metal	On-premises + Izuma support

Why Not Build It Yourself?

Consideration	Build In-House	Izuma Device Management
Time to production	Months to years	Days
Security architecture	Design from scratch	Chip-to-cloud, SOC 2 certified
Protocol support	Implement each protocol	LwM2M, CoAP, BLE, Zigbee, Modbus built in
Firmware & edge	Build pipeline + k8s integration	Delta/mesh campaigns, native k8s, SD-WAN
Factory provisioning	Custom tooling required	Factory Flow — works offline
Ongoing maintenance	Permanent engineering team	Managed service option (NoOps)

Key Differentiators: No vendor lock-in (Apache 2.0 edge stack, standard Kubernetes) · Single pane of glass across MCUs and edge servers · Factory Flow for offline provisioning · SoftBank-backed · Free tier: 200 firmware updates/month